

MODBUS RTU



PROGRAM

BASE	6 setpoint
LOAD	50 formulas
UNLOAD	50 formulas
3 PRODUCTS	50 formulas
* 6 PRODUCTS	50 formulas
* 14 PRODUCTS	50 formulas

* External 8-relay modules included

DESCRIPTION

- Weight indicator in DIN box suitable for front panel mounting.
- Dimensions: 144x72x120 mm (drilling template: 139x67 mm; 170 mm mounting depth with wirings and moderator).
- 8-digit semi-alphanumeric red LED display (14 mm height).
- 16-key keyboard with buzzer.
- IP54 front panel protection rating.
- Real-time clock/calendar with buffer battery.
- Extractable terminal boards.
- External 8-relay modules included:
 - for **6 PRODUCTS**: dimensions: 80x60x160 mm; external contacts: 115 VAC 2 A.
 - for **14 PRODUCTS**: dimensions: 80x60x160 mm, 80x60x120 mm; external contacts: 115 VAC 2 A.

INPUTS/OUTPUTS AND COMMUNICATION

- 2 independent serial ports: COM1=RS232 and COM2=RS422/485 for communication via ModBus RTU protocol, Profibus DP (RS422/485), ASCII Laumas bidirectional or one way transmission.
- 6 relay outputs controlled by the setpoint values or via protocols.
- 6 optoisolated PNP digital inputs: status reading via serial communication protocols.
- 1 load cell dedicated input.
- Current or voltage 16 bit analog output (on request).

CERTIFICATIONS



Equivalent of the CE marking for the United Kingdom

MAIN FUNCTIONS

- Connections to:
 - PLC via analog output (on request);
 - PC/PLC via COM1/2 (up to 32 instruments) with PC supervision software;
 - remote display and printer via COM1/2;
 - up to 8 load cells in parallel by junction box.
- Digital filter to reduce the effects of weight oscillation.
- Theoretical calibration (via keyboard) and real calibration (with sample weights).
- Tare weight zero setting.
- Zero tracking.
- Semi-automatic zero.
- Displaying of the maximum weight value reached (peak).
- Weight value printing with date and time via keyboard or external contact.
- Reading the load cells value expressed in mV: load cells connections continuous check.
- Password to protect the access to selected functions.

BASE PROGRAM / BASE WITH ANALOG OUTPUT

- Weight indicator with 6 setpoint.
- Hysteresis and setpoint value setting.
- Automatic zero setting at power-on.

BATCHING PROGRAM

- 50 settable formulas.
- Batching resume after a blackout.
- Automatic fall calculation.
- Tolerance error control.
- Precision batching through slow function.
- Precision batching through tapping function.
- Consumption storage.
- Print of batching data.
- Alarm contact management.
- Automatic batching.
- Batching start via external contact or keyboard.
- Signaling of minimum and maximum weight.
- Setting a single tolerance value for all formulas/products.
- Current batching can be interrupted via keyboard or external contact.
- Pause of the batching by the keyboard.

Only for:

LOAD program

- Autotare at batching start.

UNLOAD program

- Automatic loading of the product into the weighed structure.
- Management of batching with big bags.

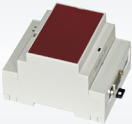
3-6-14 PRODUCTS program

- Net weight batching for each product.

TECHNICAL FEATURES

Power supply and consumption	230 (115) VAC; 50/60 Hz; 15 VA
Number of load cells • Load cells supply	up to 8 (350 Ω) - 4/6 wires • 5 VDC/120 mA
Linearity • Analog output linearity	<0.01% full scale • <0.03% full scale
Thermal drift • Analog output thermal drift	<0.0003% full scale/°C • <0.001% full scale/°C
A/D Converter	24 bit
Internal divisions	± 99999
Measurement range	± 2 mV ± 19.5 mV
Display range	± 99999 (20% $\div$ 100% full scale)
Decimals • Display increments	0-4 • x1 x2 x5 x10 x20 x50 x100
Digital filter • Readings per second	0.2-25 Hz • 5, 10, 25, 50 reading/s
Relay outputs	n. 6 - 115 VAC/30 VDC; 0.5 A
Digital inputs	n. 6
Serial ports	COM1: RS232; COM2: RS422/RS485
Baud rate	1200, 2400, 4800, 9600, 14400, 19200, 28000, 38400, 57600, 115200 (bit/s)
Analog output (on request)	16 bit. 0 $\div$ 20 mA; 4 $\div$ 20 mA (up to 300 Ω) 0 $\div$ 10 V; 0 $\div$ 5 V (min 10 k Ω)
Humidity (condensate free)	85%
Storage temperature	-20 °C + 70 °C
Working temperature	-10 °C + 50 °C

OPTIONS ON REQUEST

	DESCRIZIONE
 ANALOG OUTPUT	16 bit analog output.
	Profibus converter, IP40, suitable for Omega/DIN rail mounting, allows to connect 1 or 2 Laumas instruments in Slave configuration to any security Profibus DP Master by means of the RS422/RS485 port. Dimensions: 71x58x90 mm. 1 instrument 2 instruments

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